

Q3LOCK Local-Strict Carrier, Quartic C0, Entropy, and Gap Route Split

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This gate-level checkpoint records EXP-000804 and advances reusable result R-167 to v1.7 without allocating a new result number. The stable result identifier remains [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#). The v1.0–v1.6 weighted-energy, topology, selected-tangent, orbit-smear, and algebraic-ground-state results remain in force only with their registered boundaries.

There are two positive conclusions in this version. First, on every fixed finite ambient region, bounded local-strict, strong-star, compact-resolvent graph, and two-sided energy-constrained convergence agree. The exact onsite and commuting all-bond subflows are continuous and energy controlled on this finite-region carrier. Second, in one exact finite Gibbs system, both orientations of a configuration character have the same exact relative-entropy cost and therefore obey an evolved coordinate-tail bound. That bound decays only inverse-logarithmically.

Four route boundaries are equally part of the result. The exact unsplit full-Q3 onsite flow is not point-norm continuous on a raw momentum Weyl or basic momentum-resolvent label. The pure quartic kick does not preserve the full finite-site Buchholz–Grundling resolvent algebra. Relative entropy plus any fixed finite collection of tilted moments does not force the dynamic Gaussian history tail required by the current cutoff corridor. Finally, distinct pure disjoint ordered ground states, even with simple ground vectors, do not imply a positive broken-sector GNS gap.

The positive carrier theorem is finite-region only. It supplies no continuous-time thermodynamic split limit, all-shape exhaustion Cauchy, noncollapsing quasi-local raw oscillator algebra, phase-KMS quotient identification, or beta-independent common alpha. The norm no-go does not decide invariance of the full resolvent algebra under the unsplit flow. The entropy fixture is not a Q3LOCK dynamics counterexample. No broken-sector GNS or physical mass gap, regulator removal, continuum, physical empty-space comparison, C6, CP1, Sector A, or Pre-A closure is claimed.

2 Provenance identifiers and gate status

- Result: [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#), v1.7, R-167.
- Exploration: [EXP-000804](#).
- Closed subgate: [PA-CP1-ST8-Q3LOCK-FINITE-VOLUME-LOCAL-STRICT-ENERGY-SUBFLOW-CARRIER](#).
- Closed subgate: [PA-CP1-ST8-Q3LOCK-FIXED-GIBBS-CHARACTER-ENTROPY-TILTED-TAIL-BOUND](#).

- Open dynamics successor: [PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA](#).
- Open spectral successor: [PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY](#).
- Retained open programme gate: [PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE](#).
- Historical combined gate: [PA-CP1-ST8-Q3LOCK-QUASI-LOCAL-RAW-OSCILLATOR-ALL-EXHAUSTION-COMMON-ALPHA-AND-BROKEN-GNS-GAP](#).

The historical combined gate is *OPEN HISTORICALLY; SPLIT AND SUPERSEDED AS THE ACTIVE SUCCESSOR*. It is not closed and must not be cited as one finished theorem. This checkpoint registers exactly the following four negative results:

1. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-RAW-WEYL-BASIC-RESOLVENT-QUARTIC-POINT-NORM-CO](#);
2. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-PURE-QUARTIC-POTENTIAL-RESOLVENT-ALGEBRA-INVARIANCE](#);
3. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-ENTROPY-FINITE-MOMENT-DYNAMIC-GAUSSIAN-TAIL-INFERENCE](#);
4. [NG-2026-08-11-PRE-A-ST8-Q3LOCK-ORDERED-GROUND-DOUBLET-AUTOMATIC-GNS-GAP](#).

3 Finite-region local-strict topology

Fix a finite ambient region Y and put

$$\mathcal{H}_Y = L^2(\mathbb{R}^{8|Y|}), \quad K_Y \geq 1, \quad \mathcal{M}_Y = B(\mathcal{H}_Y) = M(K(\mathcal{H}_Y)), \quad (3.1)$$

where K_Y is any positive compact-resolvent control. On norm-bounded subsets of $\mathcal{M}_Y$, the strict topology is the strong-star topology. For one fixed $0 < s \leq 1/2$, define

$$q_{s,Y}(A) = \|AK_Y^{-s}\| + \|K_Y^{-s}A\|. \quad (3.2)$$

For $E > \inf \sigma(K_Y)$ define the two-sided energy-constrained seminorm

$$e_{E,Y}(A) = \max \left\{ \sup_{\substack{\|\psi\|=1 \\ \langle \psi, K_Y \psi \rangle \leq E}} \|A\psi\|, \sup_{\substack{\|\psi\|=1 \\ \langle \psi, K_Y \psi \rangle \leq E}} \|A^*\psi\| \right\}. \quad (3.3)$$

If $\|A\| \leq M$, spectral splitting gives the exact comparison bounds

$$e_{E,Y}(A) \leq E^s \max\{\|AK_Y^{-s}\|, \|K_Y^{-s}A\|\}, \quad (3.4)$$

$$q_{s,Y}(A) \leq 2e_{E,Y}(A) + 2ME^{-s}. \quad (3.5)$$

Since K_Y^{-s} is compact with dense range, strict convergence of a norm-bounded net is equivalent to convergence in $q_{s,Y}$. Equations (3.4)–(3.5) make this equivalent to convergence in every $e_{E,Y}$. No constant uniform in Y , no exhaustion topology, and no infinite-volume dynamics is hidden in this equivalence.

4 Exact onsite and bond subflow controls

For onsite evolution choose the separate compact-resolvent control K_Y^{os} to be a positive weighted sum of the exact zero-source onsite Hamiltonians. It commutes with the finite-region tensor onsite unitary. Onsite conjugation is therefore isometric in the versions of $q_{s,Y}$ and $e_{E,Y}$ defined by K_Y^{os} , preserves strict and strong-star topology, and has a strict- C_0 orbit on every bounded local observable.

For the exact commuting all-bond kick write

$$\beta_\delta(A) = B_\delta^* A B_\delta. \quad (4.1)$$

If A is supported in X , choose one finite ambient $Y \supset N_1(X)$ and the registered centered positive control $K_{Y,\mu}$ of R-167 v1.3. For $0 < |\delta| \leq 1$, both orientations satisfy

$$B_{\pm\delta}^* K_{Y,\mu} B_{\pm\delta} \leq M_\delta K_{Y,\mu}, \quad M_\delta = 1 + C_b |\delta|, \quad (4.2)$$

$$C_b = 1 + \frac{c^2 z^2 e^\mu}{2\chi\sqrt{\gamma}}. \quad (4.3)$$

Consequently

$$e_{E,Y}(\beta_\delta A) \leq e_{M_\delta E,Y}(A), \quad q_{s,Y}(\beta_\delta A) \leq M_\delta^s q_{s,Y}(A). \quad (4.4)$$

The support statement is exact: onsite evolution fixes X , while the bond kick sends it into $N_1(X)$ inside Y . On the common polynomial/Schwartz core, and only there when the unbounded commutators are invoked,

$$[q_x, \beta_\delta(A)] = \beta_\delta([q_x, A]), \quad (4.5)$$

$$[p_x, \beta_\delta(A)] = \beta_\delta \left([p_x, A] - \delta c \sum_{y \sim x} [q_y, A] \right). \quad (4.6)$$

Thus the directed family of finite-region multiplier algebras is a genuine beta- and phase-state-independent local-strict subflow carrier. It is a precursor, not the sought thermodynamic dynamics: split products grow in support and no pairwise all-shape exhaustion estimate has been proved. Taking a supremum over all locally normal states cannot weaken the local topology, because

$$\sup_\rho \text{Tr}(\rho D^* D) = \|D\|^2. \quad (4.7)$$

The surviving route must instead use compatible energy-density or state-tempered families across the exhaustion.

5 Full-Q3 packet no-go for point-norm continuity

At one exact eight-component Q3 site let

$$h = \sum_j \frac{p_j^2}{2\chi} + V(q), \quad K = h - \inf \sigma(h) + 1, \quad \delta = \frac{i}{\hbar} [h, \cdot], \quad (5.1)$$

and put $G = g + 3\lambda > 0$,

$$W_a = e^{-iap_0/\hbar}, \quad a \neq 0, \quad R_0 = (i + p_0)^{-1}, \quad D_a = V(q) - V(q - ae_0), \quad F_0 = \partial_0 V. \quad (5.2)$$

On Schwartz space the exact first and second derivations are

$$\delta W_a = \frac{i}{\hbar} D_a W_a, \quad (5.3)$$

$$\delta^2 W_a = \frac{i}{2\chi\hbar} \sum_j \{p_j, \partial_j D_a\} W_a - \hbar^{-2} D_a^2 W_a, \quad (5.4)$$

$$\delta R_0 = R_0 F_0 R_0, \quad (5.5)$$

$$\delta^2 R_0 = 2R_0 F_0 R_0 F_0 R_0 + \frac{1}{2\chi} R_0 \sum_j \{p_j, \partial_j F_0\} R_0. \quad (5.6)$$

The full Q3 force has leading axis coefficient Gq_0^3 . Global polynomial ellipticity with anisotropic degrees one for q and two for p gives the closed graph endpoint

$$\|\delta^2(A)K^{-3/2}\| < \infty, \quad A \in \{W_a, R_0\}. \quad (5.7)$$

The onsite propagator preserves Schwartz space, so (5.3)–(5.7) support a second-order vector Taylor formula on a common invariant core and then by graph closure. For a normalized Schwartz vector ψ , translate along the Q3 axis by

$$\psi_R = e^{-iRp_0/\hbar}\psi. \quad (5.8)$$

The exact leading limits are

$$R^{-3}\|\delta W_a \psi_R\| \longrightarrow \frac{|a|G}{\hbar}, \quad R^{-3}\|\delta R_0 \psi_R\| \longrightarrow G\|R_0^2 \psi\| > 0, \quad (5.9)$$

while

$$\|K^{3/2}\psi_R\| = O(R^6). \quad (5.10)$$

Set $t_R = \tau R^{-3}$. With

$$d_{W_a} = \frac{|a|G}{\hbar}, \quad d_{R_0} = G\|R_0^2 \psi\|, \quad C_A = \|\delta^2(A)K^{-3/2}\|,$$

and

$$M_\psi = \limsup_{R \rightarrow \infty} R^{-6} \|K^{3/2}\psi_R\| < \infty,$$

the Taylor remainder yields

$$\liminf_{R \rightarrow \infty} \|(\alpha_{\tau R^{-3}}(A) - A)\psi_R\| \geq \tau d_A - \frac{\tau^2}{2} C_A M_\psi. \quad (5.11)$$

Choose one fixed sufficiently small $\tau > 0$ for each label. The right-hand side is positive, hence

$$\liminf_{R \rightarrow \infty} \|\alpha_{\tau R^{-3}}(A) - A\| > 0. \quad (5.12)$$

Since $\tau R^{-3} \rightarrow 0$, any invariant concrete C-star algebra containing either label fails point-norm C_0 continuity for the unsplit flow. Equation (5.12) proves only a positive discontinuity. The full-flow norm jump is not an exact jump of one or two. Invariance of the full unsplit resolvent algebra remains open.

6 Pure quartic kick and resolvent-algebra scope

The pure potential subflow has a stronger exact boundary. Along one Q3 axis,

$$W_4(re_0) = \frac{G}{4}r^4, \quad U_t = e^{itW_4(Q)/\hbar}, \quad R_\mu = (i\mu - p_0)^{-1}, \quad A_t = U_t R_\mu U_t^*. \quad (6.1)$$

For the configuration translation $S_s = e^{-isp_0/\hbar}$, a normalized translated-dilated packet of width $R^{1/2}$ has momentum spread $O(R^{-1/2})$. After multiplication by the relative quartic phase, its momentum center is

$$k_R = 3tGsR^2 + O(R), \quad \|(p_0 - k_R)\chi_R\| = O(R^{3/2}) = o(|k_R|). \quad (6.2)$$

One resolvent action tends to $(i\mu)^{-1}$ and the other to zero. The Cayley-transform upper bound matches the packet lower bound, so for every nonzero s, t ,

$$\|S_s A_t S_s^* - A_t\| = \frac{1}{|\mu|}. \quad (6.3)$$

Elements of the finite-dimensional field/resolvent algebra have norm-continuous Weyl-conjugation orbits on finite-dimensional phase-space subspaces; see Georgescu–Iftimovici, Proposition 2.38(2), <https://arxiv.org/abs/1902.10026>. Therefore

$$U_t R_\mu U_t^* \notin \mathcal{R}(\mathbb{R}^{16}, \sigma), \quad t \neq 0. \quad (6.4)$$

This resolvent algebra is unital, so its multiplier strict topology equals its norm topology. Strict continuity cannot repair the pure-kick route. Equation (6.4) blocks a Trotter construction that requires the kinetic and pure quartic subflows separately to be internal automorphisms. It neither proves nor disproves invariance under the unsplit kinetic-plus-quartic flow.

7 Two-sided fixed-Gibbs character entropy

Let $\rho = Z^{-1}e^{-\beta H}$ be one exact finite-volume Q3LOCK Gibbs density and let $A = W_\xi = e^{i\xi \cdot q}$. The exact momentum shift is

$$A^* H A - H = \frac{\hbar}{\chi} \xi \cdot p + \frac{\hbar^2}{2\chi} \|\xi\|^2. \quad (7.1)$$

The real Hamiltonian is time-reversal invariant, so the Gibbs momentum mean vanishes. Unitary covariance of logarithms then gives both orientations,

$$S(A\rho A^*|\rho) = S(A^*\rho A|\rho) = \frac{\beta\hbar^2}{2\chi} \|\xi\|^2 =: S_\xi. \quad (7.2)$$

Gibbs invariance makes (7.2) valid also for $A_t = \alpha_t(A)$. For a coordinate-tail projection E , put

$$p = \text{Tr}(\rho E), \quad q_+(t) = \text{Tr}(A_t \rho A_t^* E), \quad q_-(t) = \text{Tr}(A_t^* \rho A_t E). \quad (7.3)$$

Binary data processing gives

$$S_\xi \geq d_{\text{bin}}(q_\pm(t)||p) \geq q_\pm(t) \log(1/p) - \log 2, \quad (7.4)$$

and hence

$$q_\pm(t) \leq \min \left\{ 1, \frac{S_\xi + \log 2}{\log(1/p)} \right\}. \quad (7.5)$$

If the exact coordinate-tail theorem supplies

$$p \leq M_a |S| e^{-aL^2}, \quad aL^2 > \log(M_a |S|), \quad (7.6)$$

then

$$q_{\pm}(t) \leq \min \left\{ 1, \frac{S_{\xi} + \log 2}{aL^2 - \log(M_a |S|)} \right\}. \quad (7.7)$$

This is an exact finite-Gibbs, evolved-character, two-orientation theorem. Under (7.6) it is only $O(L^{-2})$, not a Gaussian history tail.

8 Entropy and finite moments do not force a Gaussian tail

For $n \geq 2$ and any fixed integer $m \geq 3$ on $\mathbb{C}^2$, let

$$q_n = nP_1, \quad H_n = \frac{n^4}{\beta} P_1, \quad \rho_n = \frac{\text{diag}(1, e^{-n^4})}{1 + e^{-n^4}}. \quad (8.1)$$

Write $\Delta_n = p_{0,n} - p_{1,n} = \tanh(n^4/2)$ and choose

$$\sin^2 \theta_{n,m} = \frac{1}{\Delta_n n^{2m}}, \quad U_{n,m}^{\pm} = e^{\mp i \theta_{n,m} \sigma_y}, \quad \rho_{n,m}^{\pm} = U_{n,m}^{\pm} \rho_n U_{n,m}^{\pm*}. \quad (8.2)$$

Exact two-by-two arithmetic gives, for both signs,

$$\text{Tr}(\rho_{n,m}^{\pm} P_1) = p_{1,n} + n^{-2m}, \quad (8.3)$$

$$S(\rho_{n,m}^{\pm} \| \rho_n) = n^{4-2m}, \quad \text{Tr}[(\rho_{n,m}^{\pm} - \rho_n) H_n] = \frac{n^{4-2m}}{\beta}, \quad (8.4)$$

$$\text{Tr}(\rho_{n,m}^{\pm} q_n^r) = n^r p_{1,n} + n^{r-2m}, \quad 0 < r \leq 2m. \quad (8.5)$$

For $r = 0$ the moment is exactly one. The reference Gibbs state has every Gaussian coefficient,

$$\text{Tr}(\rho_n e^{a q_n^2}) \leq 1 + e^{a^2/4}, \quad a > 0. \quad (8.6)$$

For any fixed $T > 0$, the two rotations are exact time- T flows of the bounded auxiliary drives

$$K_{n,m}^{\pm} = \pm \frac{\hbar \theta_{n,m}}{T} \sigma_y. \quad (8.7)$$

At $m = 4$ the tail is exactly $p_{1,n} + n^{-8}$, the relative entropy is n^{-4} , the energy excess is $1/(\beta n^4)$, and

$$\text{Tr}(\rho_{n,4}^{\pm} q_n^8) = 1 + n^8 p_{1,n} \leq 1 + 4e^{-2}. \quad (8.8)$$

More generally, after any finite moment ceiling is preregistered, one can fix m above it. A genuine Gibbs reference, all-coefficient Gaussian reference tails, vanishing entropy and energy excess, both orientations, and all the requested finite tilted moments still allow a polynomial tail. This fixture rejects that inference only. Its auxiliary drive is not the equilibrium flow of H_n , and it does not rule out a stronger Q3LOCK orbit quasi-invariance estimate.

9 The remaining history corridor

For the bounded coordinate cutoff, the safe interaction corridor is

$$B_{R,L}(T) = 8\sqrt{2}|X|||A||e^{\nu_L T} \frac{(\nu_L T)^R}{R!}, \quad \nu_L \leq \frac{96cL^2}{\hbar}. \quad (9.1)$$

Choosing $L = R^\alpha$ makes (9.1) vanish on compact time intervals for $0 < \alpha < 1/2$, but the estimate pays e^{CL^2} . Neither the inverse-logarithmic bound (7.7) nor any fixed finite-moment version of (8.3)–(8.8) absorbs that loss. A Trotter telescope requires uniform control for every partial history P and its adjoint history,

$$\phi_0(P^*EP) + \phi_0(PEP^*), \quad (9.2)$$

together with the corresponding $(V - V_L)^2$ -weighted expressions. Static tails and one-step energy control do not prove (9.2). The next dynamics gate must establish a compatible two-orientation history estimate across all exhaustions, or replace the corridor with a state-weighted all-bond resummation that avoids its exponential loss.

10 Ordered ground states do not imply a GNS gap

Let

$$\mathcal{K}_0 = \mathbb{C}\Omega \oplus L^2((0,1), dx), \quad h_0\Omega = 0, \quad (h_0f)(x) = xf(x), \quad (10.1)$$

and define

$$\mathcal{A} = B(\mathcal{K}_0) \oplus B(\mathcal{K}_0), \quad \alpha_t = \text{Ad}(e^{ith_0/\hbar}) \oplus \text{Ad}(e^{ith_0/\hbar}). \quad (10.2)$$

Because h_0 is bounded with norm one, this is a point-norm C_0 C-star dynamics. The two component vector states

$$\omega_+(A \oplus B) = \langle \Omega, A\Omega \rangle, \quad \omega_-(A \oplus B) = \langle \Omega, B\Omega \rangle \quad (10.3)$$

are pure, disjoint, exact ground states. Summand-swap parity exchanges them, the central order observable $Z = (1, -1)$ separates them, and each GNS ground vector is simple. Nevertheless

$$\sigma(h_0) = \{0\} \cup [0, 1], \quad \inf(\sigma(h_0) \setminus \{0\}) = 0. \quad (10.4)$$

The positive implementing spectrum accumulates at zero. With the repository convention $\delta = (i/\hbar)[H, \cdot]$, a genuine sector gap requires an independent coercive estimate on a dense invariant core,

$$-i\hbar\omega(A^*\delta(A)) \geq \Delta\{\omega(A^*A) - |\omega(A)|^2\}, \quad \Delta > 0, \quad (10.5)$$

separately in both selected broken-sector GNS representations. Neither distinct order nor the v1.6 ground-state construction proves (10.5).

11 Gate decision

The finite-region local-strict energy subflow carrier and the fixed-Gibbs character entropy tail are closed exactly in their stated finite scopes. The former supplies the correct local bounded topology and exact subflow controls; the latter supplies a real but too-weak two-orientation tail.

The active dynamics successor is [PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA](#). It must close (9.2), all-shape exhaustion Cauchy, continuous-time group completion, a noncollapsing spatial algebra, the zero-source Hamiltonian generator, and both phase-KMS quotient identifications. The independent spectral successor is [PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY](#); it must prove (10.5) after the broken ground representations have been identified on that common algebra. Neither successor implies the other.

12 Devil’s-advocate review

1. **DISMISSED:** strict topology on the unital full resolvent algebra cannot repair the pure-kick norm jump. Its multiplier strict topology is norm. The finite-region $B(\mathcal{H}_Y) = M(K(\mathcal{H}_Y))$ carrier is a different local multiplier construction.
2. **UPHELD as a boundary:** the local-strict equivalence has no uniformity in Y . Growing support and the missing two-orientation history estimate prevent its promotion to a thermodynamic dynamics.
3. **DISMISSED:** the packet argument is not only a formal first derivative. The $K^{3/2}$ graph endpoint controls the exact second-order remainder at $t_R = \tau R^{-3}$.
4. **UPHELD as an overreach:** (5.12) gives a positive liminf only, not an exact jump of one or two. Pure-kick non-invariance also does not decide unsplit resolvent-algebra invariance.
5. **DISMISSED:** data processing does not supply the Gaussian tail. It gives only (7.7), and the general- m fixture preserves arbitrary preregistered finite moments while retaining a polynomial tail.
6. **UPHELD as a boundary:** the two-level auxiliary rotations are not Q3LOCK equilibrium flows. They falsify only the entropy-plus-finite-moment inference, not the existence of a stronger Q3LOCK estimate.
7. **DISMISSED:** pure disjoint ordered ground states with a simple ground vector need not be gapped. The bounded exact fixture retains all those properties while its positive implementing spectrum accumulates at zero.
8. **DISMISSED:** the unbounded bond commutator recurrences are not claimed on all of $B(\mathcal{H}_Y)$. Equations (4.5)–(4.6) are explicitly restricted to their common polynomial/Schwartz core.

13 Reproduction

The three exact v1.7 scripts are [codes/foundations/pre_a_cp1_st8_q3lock_local_strict_quartic_c0_entropy_gap_route_split.py](#), [codes/foundations/pre_a_cp1_st8_q3lock_local_strict_quartic_c0_entropy_gap_route_split_independent.py](#), and [codes/foundations/pre_a_cp1_st8_q3lock_local_strict_quartic_c0_entropy_gap_route_split_verify.py](#). From the repository root, run them in this order:

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$stem = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_strict_quartic_c0_entropy_gap_route_split'
)
& $py -X utf8 ($stem + '.py')
& $py -X utf8 ($stem + '_independent.py')
& $py -X utf8 ($stem + '_verify.py')
```

The expected proof-layer outputs are primary PASS 189/189, independent PASS 109/109, and integrated PASS 153/153. The synthesis source is complete. Its sibling PDF is governed by the final manifest hash, page-count, freshness, and all-page visual-QA fields.

14 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v1.7; no new result number.
Exploration:	EXP-000804.
Precise statement:	Finite-region bounded local-strict, strong-star, compact-resolvent graph and two-sided energy topologies agree; exact onsite and all-bond subflows are controlled there. Fixed finite-Gibbs characters have an exact two-orientation entropy cost and inverse-logarithmic evolved-tail bound.
Route boundaries:	Unsplit full-Q3 raw momentum labels are not point-norm C0; the pure quartic kick does not preserve the full resolvent algebra; entropy plus fixed finite moments does not force a Gaussian history tail; ordered ground sectors do not imply a GNS gap.
Scope:	T0 exact finite-region and fixed-finite-Gibbs theorems plus four independently scoped no-go results; claim_bearing false.
Dependencies:	EXP-000792, EXP-000799, EXP-000803, EXP-000804; R-167 v1.0--v1.6; exact finite Q3 Hamiltonians.
Evidence grade:	ANALYTIC + EXACT + primary + non-importing independent + integrated verification.
Reproduction command:	Run the three commands under Reproduction in the listed order, with the integrated verifier last.
Expected output:	Primary PASS 189/189; independent PASS 109/109; integrated PASS 153/153.
Checkpoint PDF status:	One gate-level source/PDF pair governed by final hash, page-count, freshness, and visual-QA checks.
Falsification gate:	Any topology-equivalence, subflow-form, packet scaling, positive-liminf, exact kick norm, entropy orientation, fixture moment, ground-spectrum, provenance, hash, count or scope mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	No thermodynamic split limit. No all-exhaustion common alpha or quasi-local raw oscillator algebra, unsplit resolvent-algebra invariance verdict, phase-KMS quotient, broken-sector GNS or physical gap, regulator removal, continuum, physical-empty sign, C6, CP1, Sector A or Pre-A closure.
Next required action:	Prove uniform two-orientation partial-history control and all-shape exhaustion common alpha; separately prove sectorwise GNS coercivity after identifying the broken representations.